Firewall

- A firewall is like a sentry(guard) keep monitoring and control the accessibility (both inside and outside the network)
- Firewall is a ***security barrier*** between two networks that screens traffic coming in and out of the gate of one network to accept or reject connections and services according to a set of rules.

Firewall

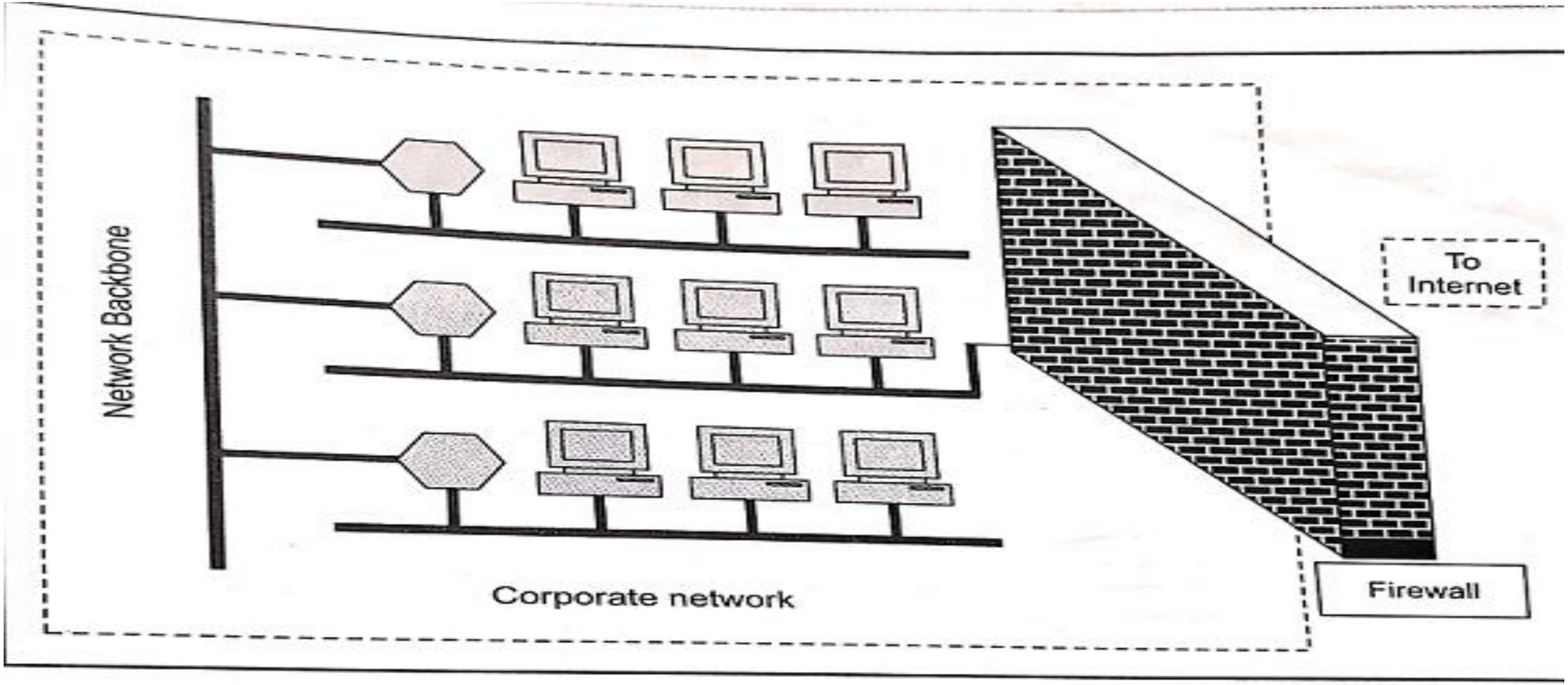


Fig. 9.6 *Firewall*

Firewall

Characteristics:

- All traffic from inside to outside and vice versa must pass through the firewall.
- Only the traffic authorized as per the local security policy should be allowed to pass through.
- The firewall itself must be strong enough

Types of Firewall:

1. Packet Filtering Firewall
2. Application Gateway
3. Circuit Level Gateway

Packet Filtering Firewall

- **Packet –** A set of rules to each packet & on outcome decides to forward/discard the packet. Also called as Screening Router/Filter.

A packet filter performs following functions
Receive each packet arrives
Pass them through set of rules
If no rule match, take default action(*discard/ accept Packets*)

Advantage *is simplicity, fast in their operating speed*
Disadvantage *is difficult in setting packet filter rules, lack of support for Authentication.*

Attacker try to break security of a packet filter by **IP address spoofing, Source routing attacks and Tiny fragment attacks** techniques.

Packet Filtering Firewall

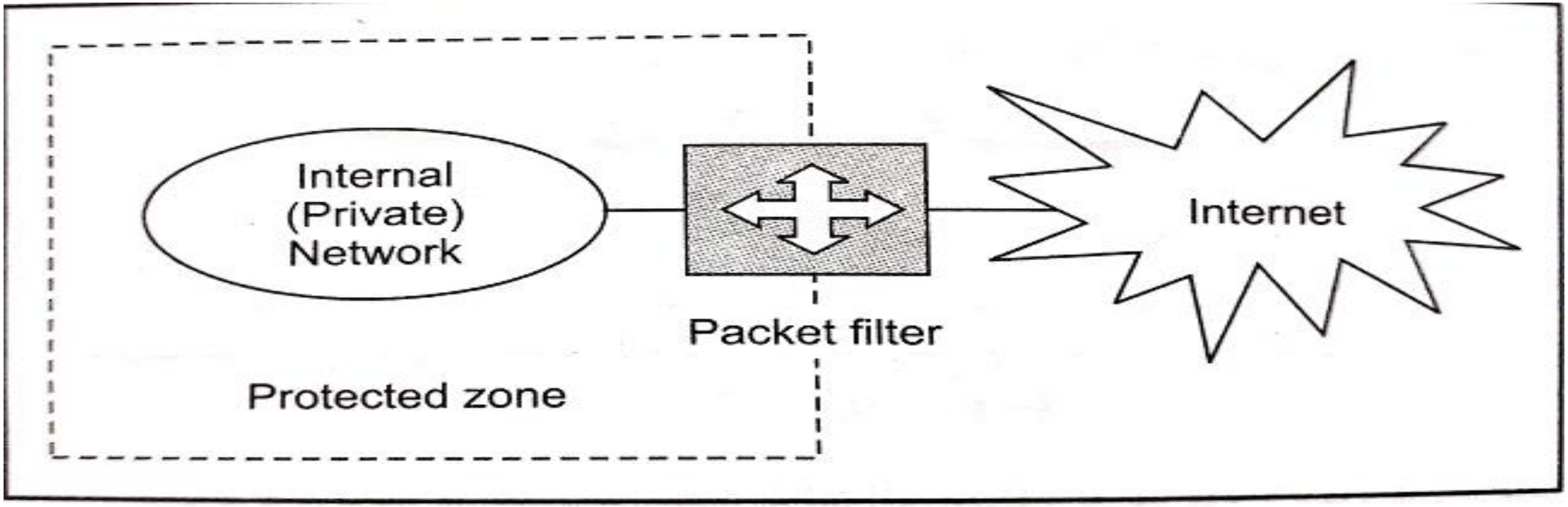


Fig. 9.8 *Packet filter*

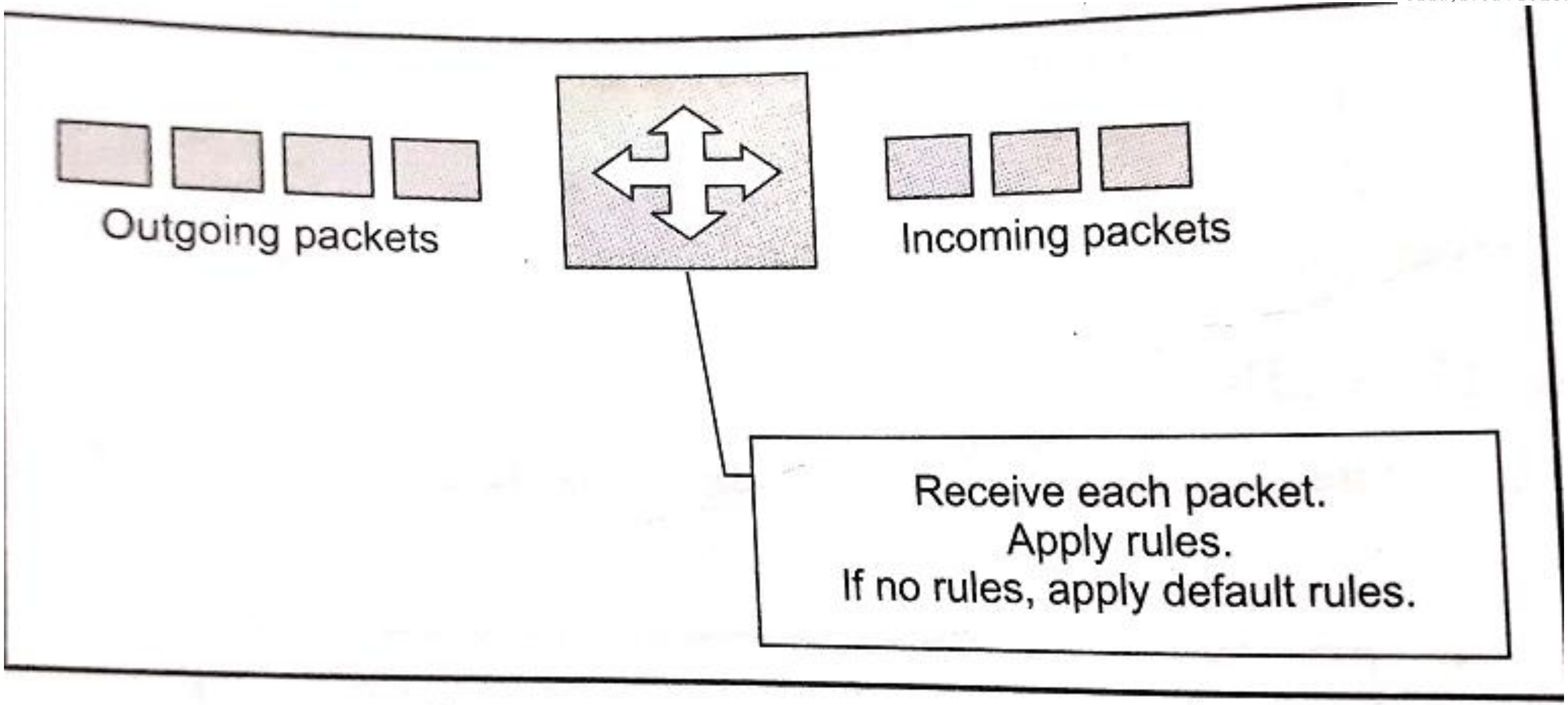
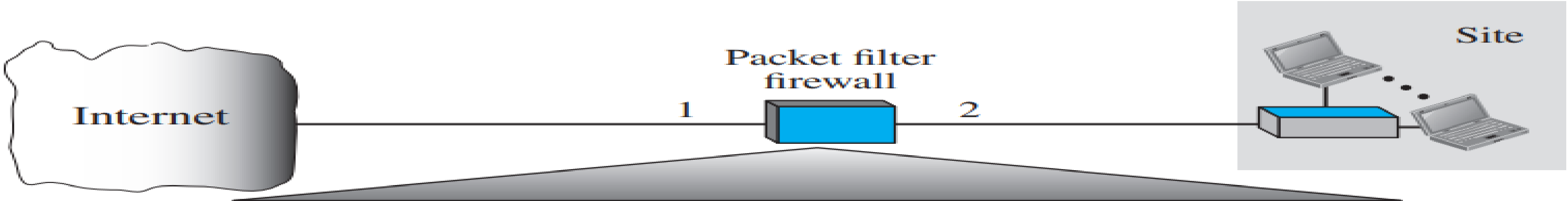


Fig. 9.9 *Packet filter operation*



Interface	Source IP	Source port	Destination IP	Destination port
1	131.34.0.0	*	*	*
1	*	*	*	23
1	*	*	194.78.20.8	*
2	*	*	*	80

1. Incoming packets from network 131.34.0.0 are blocked (security precaution). Note that the * (asterisk) means "any."
2. Incoming packets destined for any internal TELNET server (port 23) are blocked.
3. Incoming packets destined for internal host 194.78.20.8 are blocked. The organization wants this host for internal use only.
4. Outgoing packets destined for an HTTP server (port 80) are blocked. The organization does not want employees to browse the Internet.

Issues:

- IP address Spoofing
- Source Routing attack
- Tiny fragment attacks

IP address Spoofing

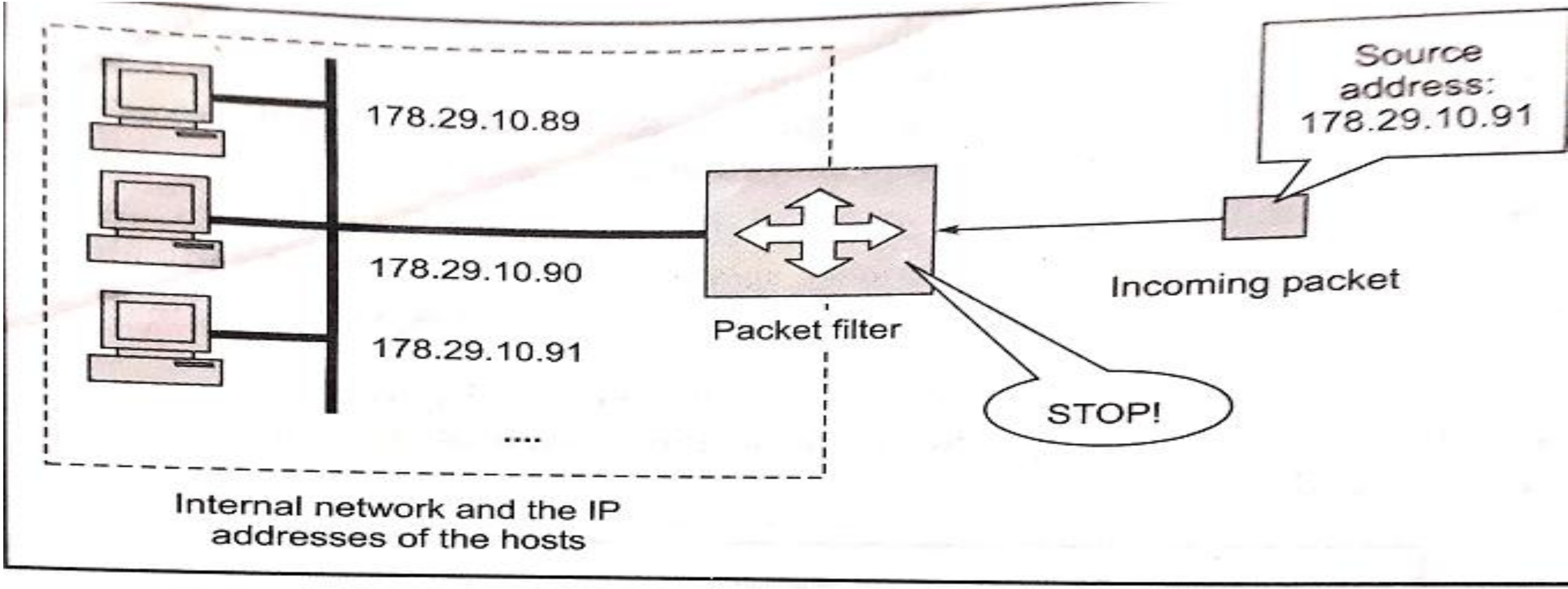


Fig. 9.11 *Packet filter defeating the IP address spoofing attack*

Application Gateway

- **Application gateway –** is also called as **proxy server,** because it acts like a substitute and decide about the flow of application level traffic. It also called as **bastion server**
- It works as follows
  - Internal user contact application gateway using TCP/IP application, such as HTTP or TELNET
  - Application gateway asks the user about the remote host detail, user id and password to access
  - User provide those information to application gateway
  - Application gateway passes the user's packets to the remote host

Application Gateway

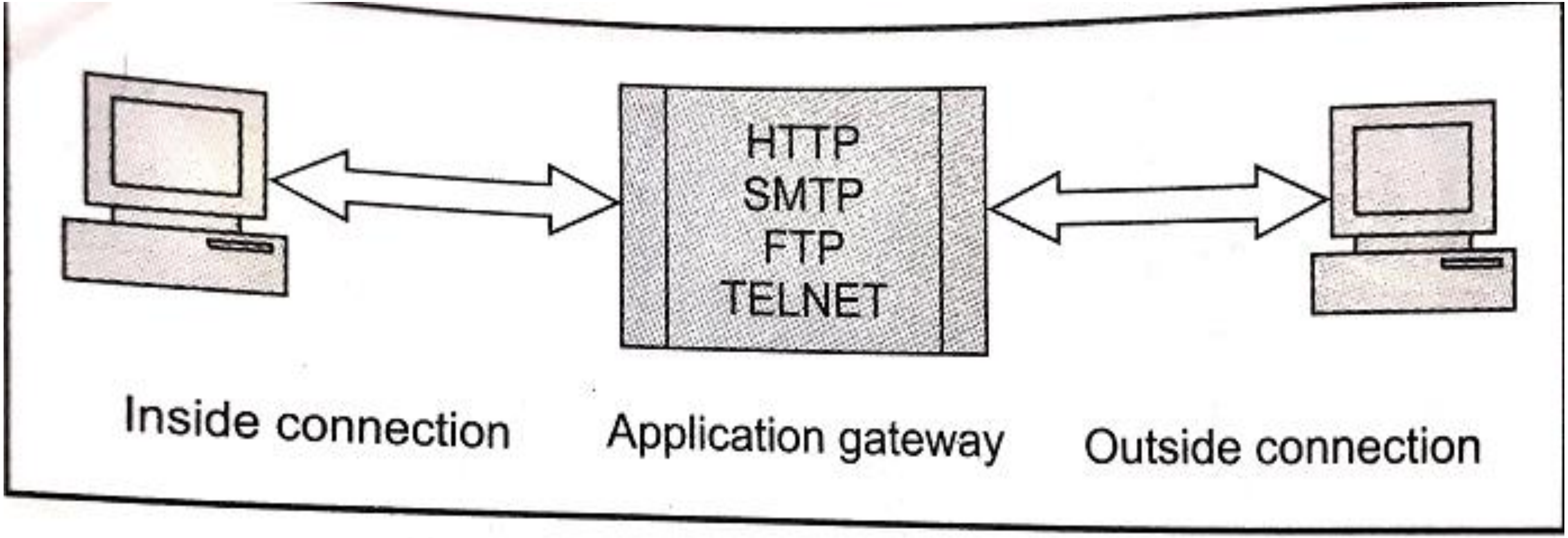


Fig. 9.13 *Application gateway*

Circuit Level Gateway

- will create new connection between itself and remote host

– It change the source IP address in the packets, so internal user’s IP are hidden from outside world

Circuit Level Gateway

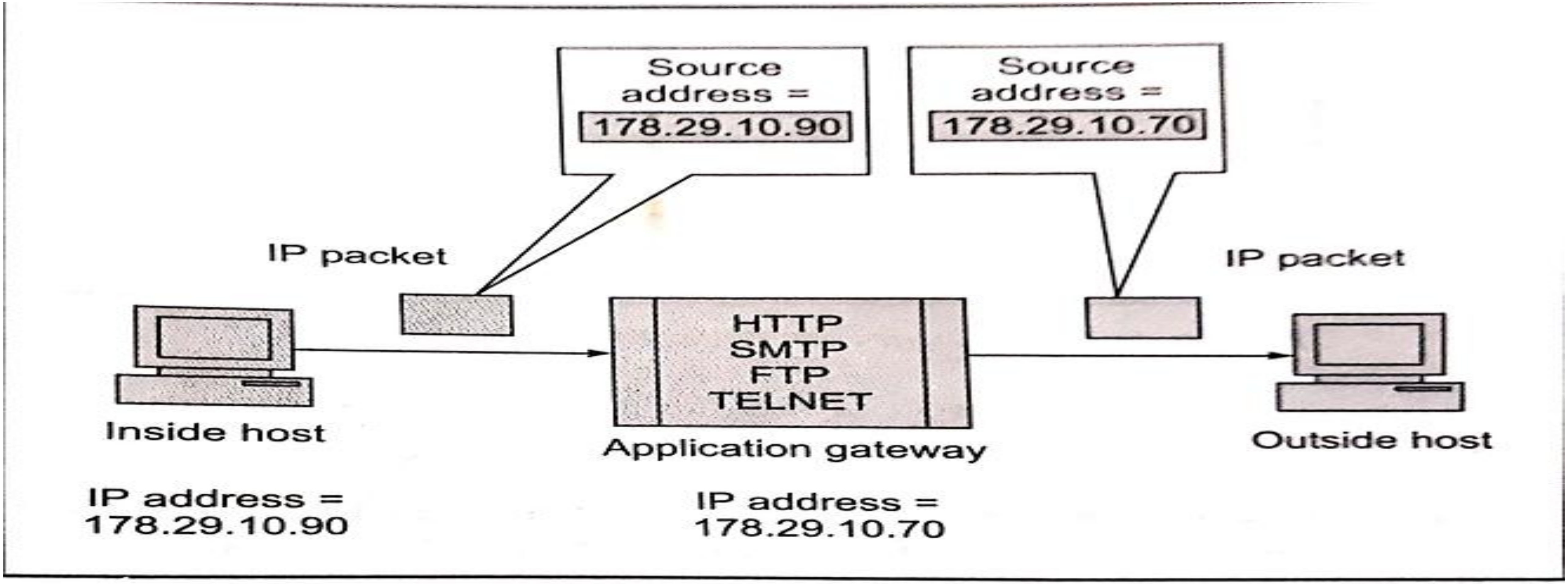


Fig. 9.14 *Circuit gateway operation*

Firewall Configurations

- Screened Host Firewall – Single Homed Bastion
- Screened Host Firewall – Dual Homed Bastion
- Screened Subnet Firewall

Screened Host Firewall – Single Homed Bastion

- 1. It consists of two parts – packet-filtering router and application gateway
- The packet filter ensures the incoming traffic is allowed, if it is destined for application gateway by examine destination address of incoming IP packet. And for outgoing packet, the source address is examined
- The application gateway performs authentication and proxy function
- This configuration increases security at both packet and application levels , it gives flexibility to network
- A disadvantage here is if packet filter is attacked then its security compromised and whole network is attacked

Screened Host Firewall – Single Homed Bastion

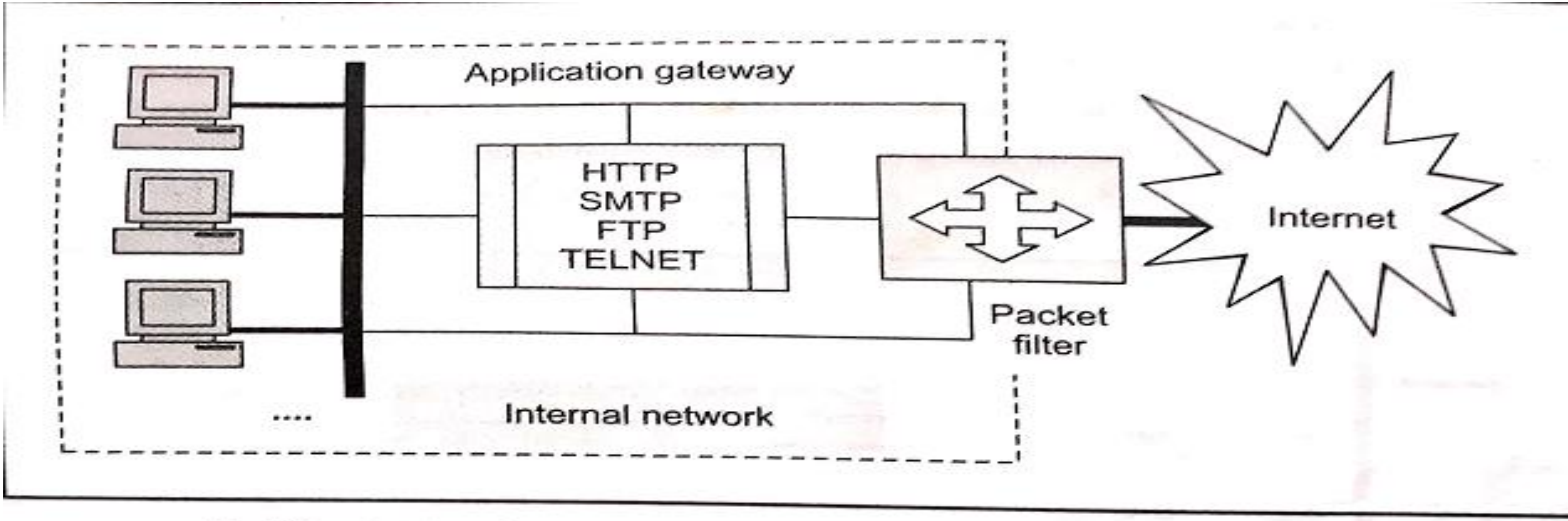


Fig. 9.21 *Screened host firewall, Single-homed bastion*

Screened Host Firewall – Dual Homed Bastion

To overcome earlier configuration, direct connection between internal hosts and packet filter are avoided.

It connects only to application gateway, has a separate connection with internal hosts, if packet filter is attacked the application gateway only visible not the internal hosts.

Screened Host Firewall – Dual Homed Bastion

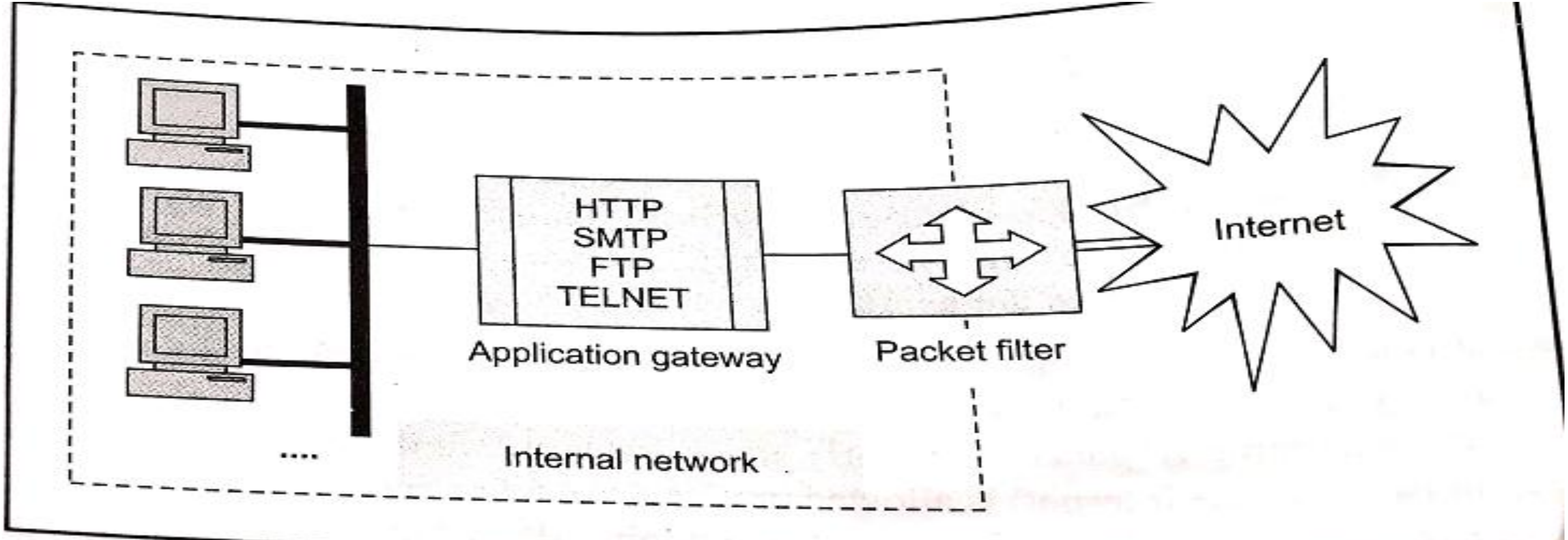


Fig. 9.22 *Screened host firewall, Dual-homed bastion*

Screened subnet Firewall

- It offers the highest security among the possible firewall configuration.
- Here two packet filters are used, one between internet and application gateway, another one between application gateway and internal network

Screened subnet Firewall

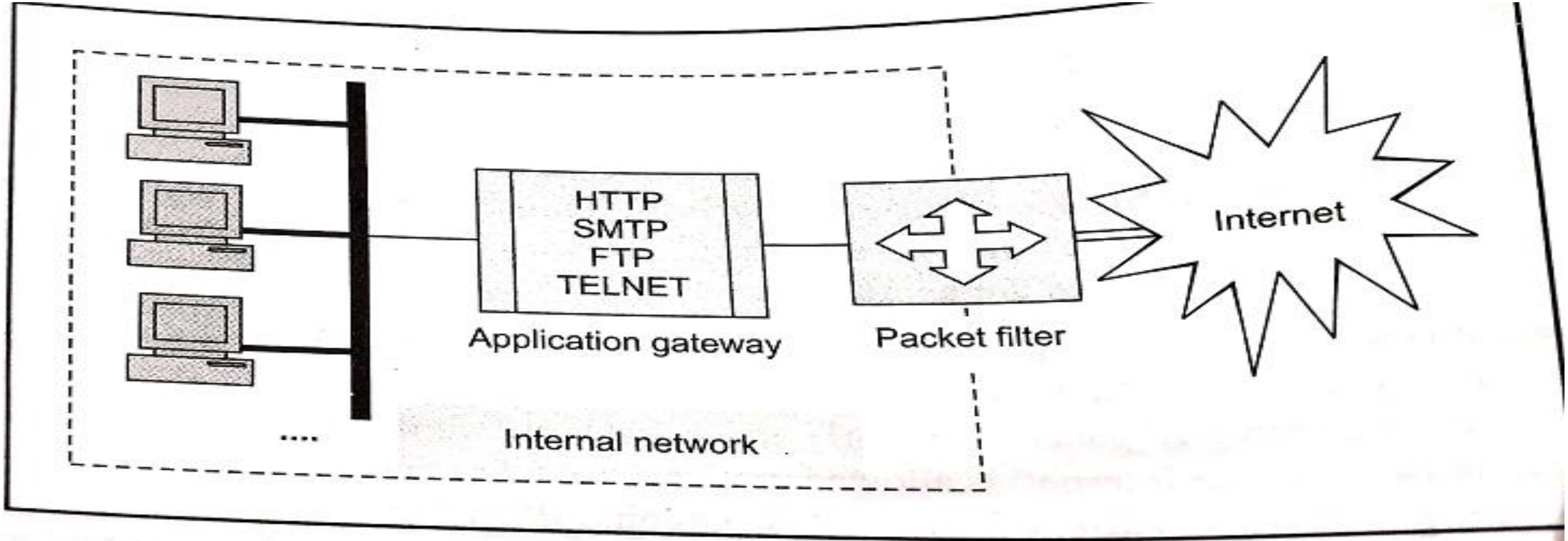


Fig. 9.22 *Screened host firewall, Dual-homed bastion*

Limitations of Firewall

- ## Insider's intrusion

  - Firewalls mainly guard against **external attacks**.

  - If an attack comes from **inside the network (insiders, employees, compromised devices)**, it may not be blocked.

  Example: An employee accessing or leaking sensitive data.

- ## Direct Internet traffic

  - If users connect to the internet **outside the firewall** (e.g., mobile hotspot, VPN), security is bypassed.

  Example: A laptop using public Wi-Fi instead of the company network.

- ## Virus attacks

- Malware hidden in:
  - Emails
  - Downloads
  - Encrypted traffic

  may pass through.

Example: Virus inside an email attachment.